Objectives:

- Install Oracle and SQL Server Databases
- Configure Oracle and SQL Server Databases
- Start-up a database instance/service
- Shutdown a database instance/service
- Troubleshoot common database connectivity issues
- Create database system network configuration settings
- Connect to a database using various client tools and methods

Tasks/Activities:

1. Oracle Database Installation, Configuration and Connection

1.1. Install Oracle (latest version)

- Download Oracle database installation files
- Unzip files
- Run setup.exe to launch installation wizard
- Make a note of the installation options selected in your lab report
- Note any challenges you encountered and how you resolved them

1.2. Configure Oracle Database

- Create a database using Oracle Database Configuration Assistant (NB: You may have already done this when you installed the database. If you did, then you can skip this task)

1.3. Connect to Oracle database using SQL Plus

Use SQLPlus (CLI) to connect to the Oracle database using the System password you created during the installation/configuration

- Did you encounter any issues? How did you resolve them?
- Make a note of them in your Lab report
- Make a note of the correct statement you used to successfully connect to the database

1.4. Connect to Oracle database using SQL Developer

Use SQL Developer (IDE) to connect to the Oracle database

- Did you encounter any issues? How did you resolve them?

- Make a note of them in your Lab report
- Make a note of the correct settings and values you used to successfully connect to the database e.g. Connection type

1.5. Configure Oracle Net Service

- Do some research and explain the purpose of the TNSNames.ora file. Include this in your lab report
- Locate your TNSNames.ora file and copy the content to your lab report

1.6. Shutdown Oracle database instance

- Make note of the command(s) used to shut down your Oracle database instance
- Attempt to connect to your database after the instance is shutdown. What error message did you receive? Make a note in your lab report

1.7. Startup Oracle database instance

- Make a note of the command(s) used to startup your Oracle database instance
- Connect to your database via SQLPlus to confirm it is active
- Did you encounter any errors? How did you resolve them? Make a note in your lab report

2. Microsoft SQL Server Installation, Configuration and Connection

2.1. Install SQL Server (latest version)

- Download SQL Server installation file
- Mount *.ISO file
- Run setup.exe to launch installation wizard
- Make a note of the installation options selected in your lab manual
- Note any challenges you encountered and how you resolved them

2.2. Connect to SQL Server Database Instance using SQL Server Management Studio (SSMS)

2.3. Configure SQL Server Database

- Create a new database in SQL Server Management Studio

- Generate the database creation script and add this to your lab manual

2.4. Connect to SQL Server database using SQL Server Management Studio

- Did you encounter any issues? How did you resolve them?
- Make a note of them in your Lab report
- Make a note of the correct variable you used to successfully connect to the database

2.5. Shutdown the SQL Server database service

- Which method did you use to shutdown the service? Make a note in your lab report
- Attempt to connect to the database while it is offline. What happened? Make a note in your lab report. Copy and paste the error message.

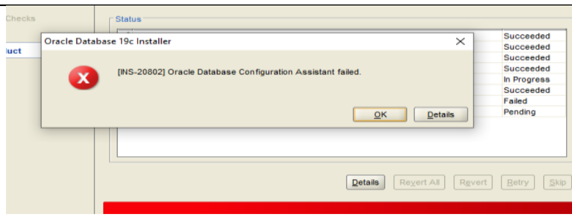
2.6. Start-up the SQL Server database service

- Use a different method from the one you used to shutdown the database. Which method did you use? Make a note in your lab report.
- Confirm that the database service is running by connecting to the database. Did you receive any errors? How did you resolve them? Make a note in your lab report

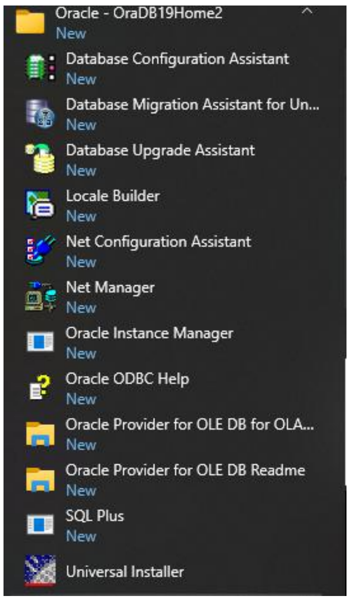
2.7. Configure Network connectivity for your Database

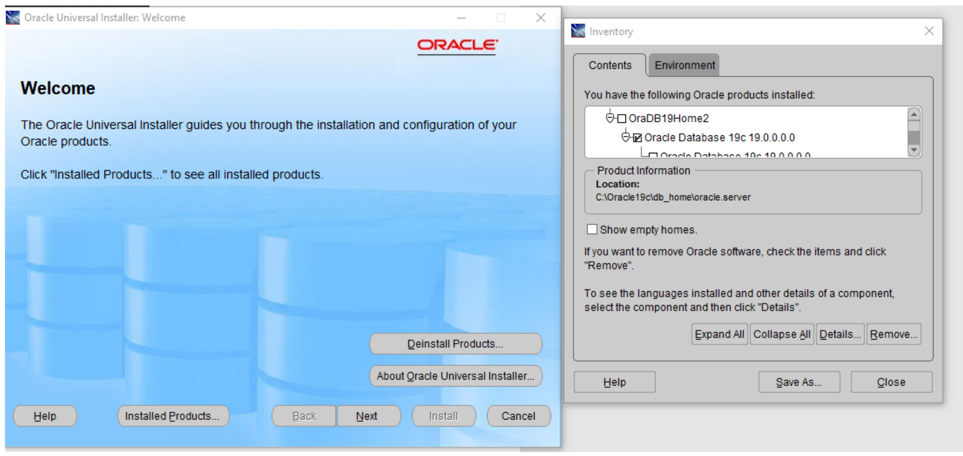
- Windows Firewall - Open port 1433 to incoming connections
- TCP/IP Protocols - Use SQL Server

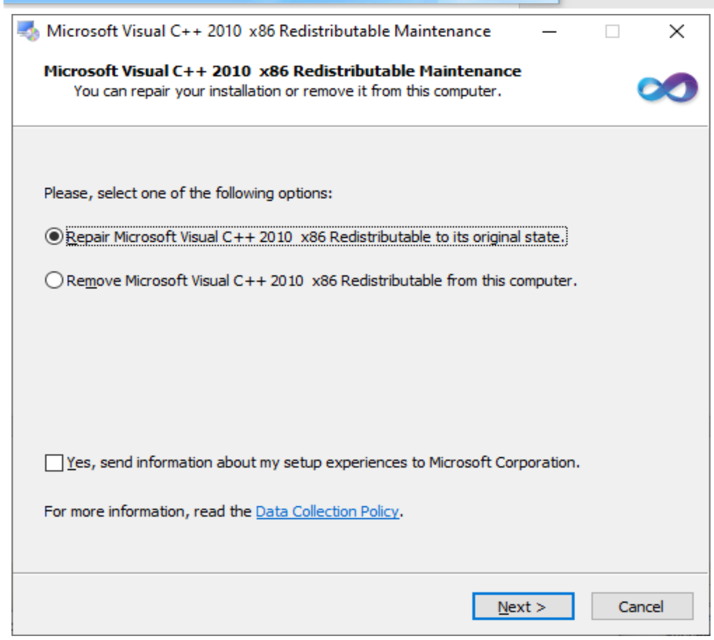
Lab Report

Difficulties Encountered		
#	Issue	Resolution
1	After unpacking the Oracle install package, and double clicking on "setup.exe" wasn't running or opening.	Change the offending folder name by removing any spaces embedded in the name and create a new local account with no space user folder. Then re-run the installer.
2	 "[INS-20802] Oracle Net Configuration Assistant failed." Error at install product step.	Navigate to the new Oracle folder in start button list and open the application "Universal Installer" and select "deinstall products" button and check "Oracle Database 19c…" and select the remove button to uninstall. Download/Install or repair "Microsoft Visual C++ 2010 x86 Redistributable Maintenance". Link: https://www.microsoft.com/en-us/download/confirmation.aspx?id=26999
3	I was unable to locate the SQL Developer application	Had to manually search the internet for the SQL Developer application developer which was found on Oracle website. Link : https://www.oracle.com/tools/downloads/sqldev-downloads.html Where I had to sign in to an account to be able to download.

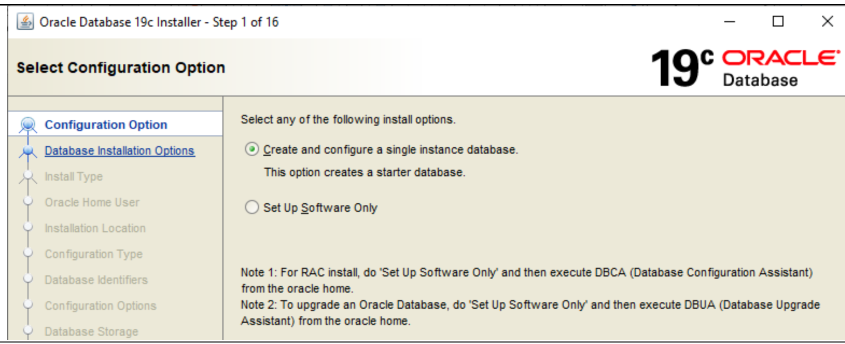
#2 GUI Resolution







Selected Configuration Options (ORACLE)

<table>
<tr><th>STEP</th><th>Selection (GUI view)</th></tr>
<tr><td>1</td><td>
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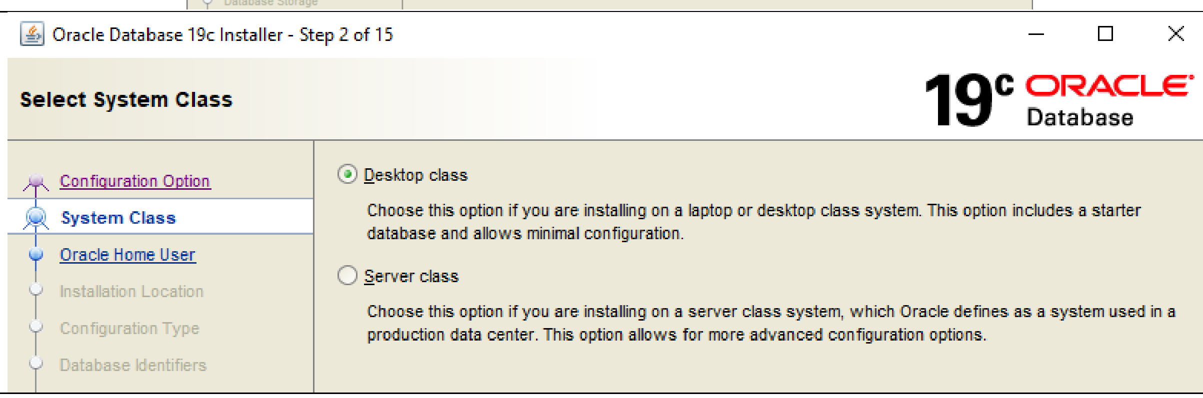
<tr><td>2</td><td>
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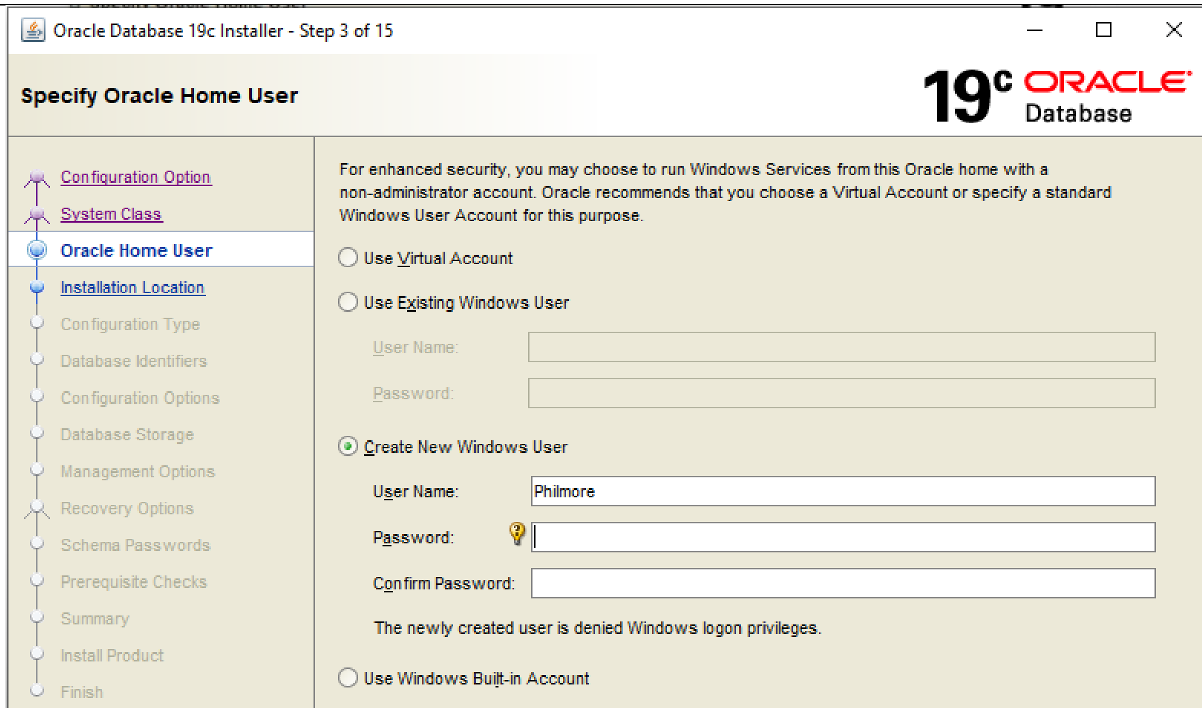
<tr><td>3</td><td>
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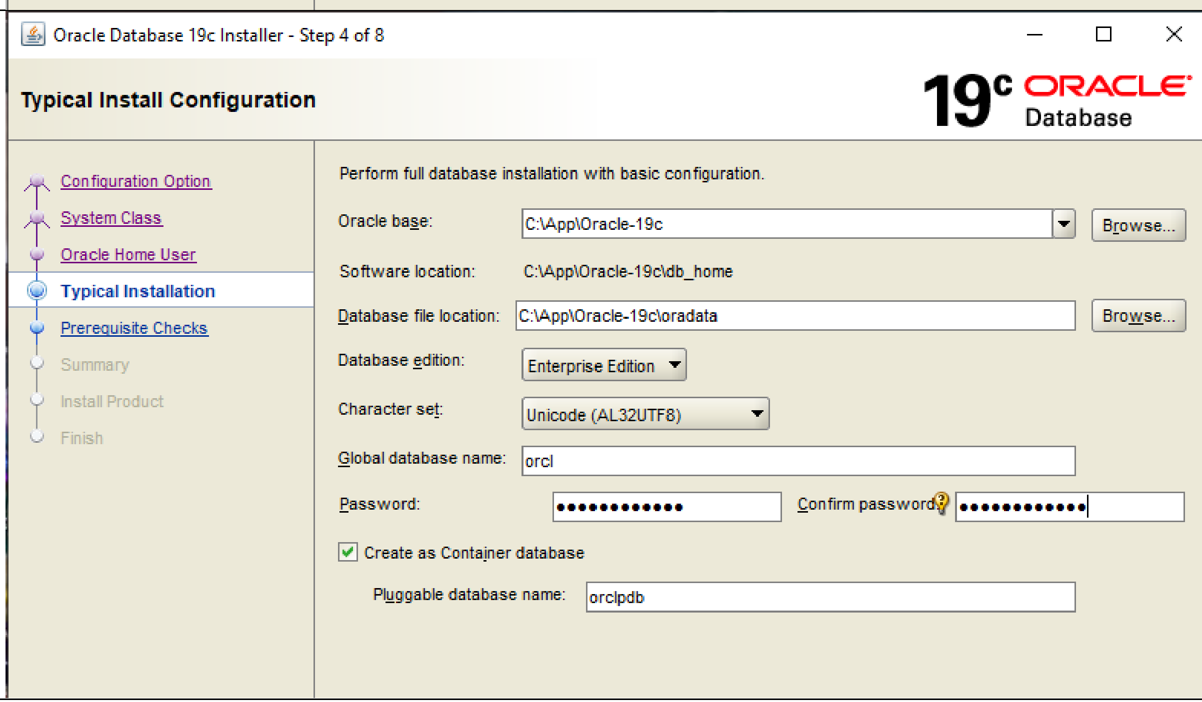
<tr><td>4</td><td>
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<tr><td>5</td><td>Prerequisites Checks</td></tr>
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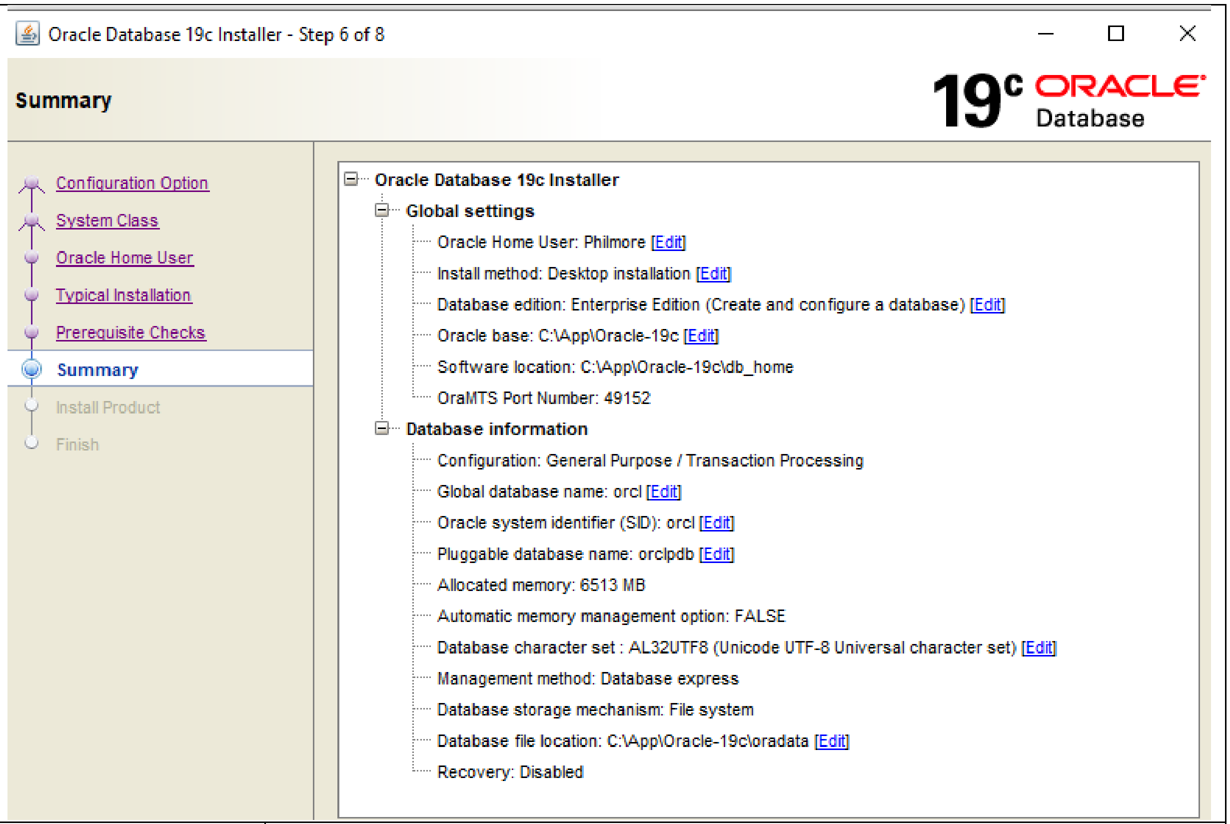
<tr><td>6</td><td colspan="2">
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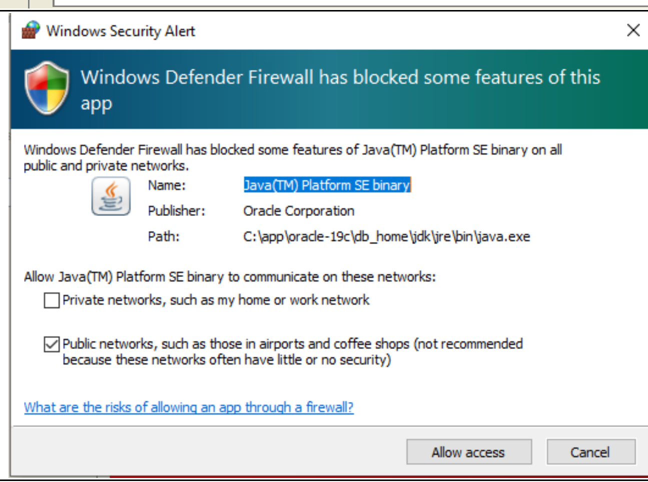
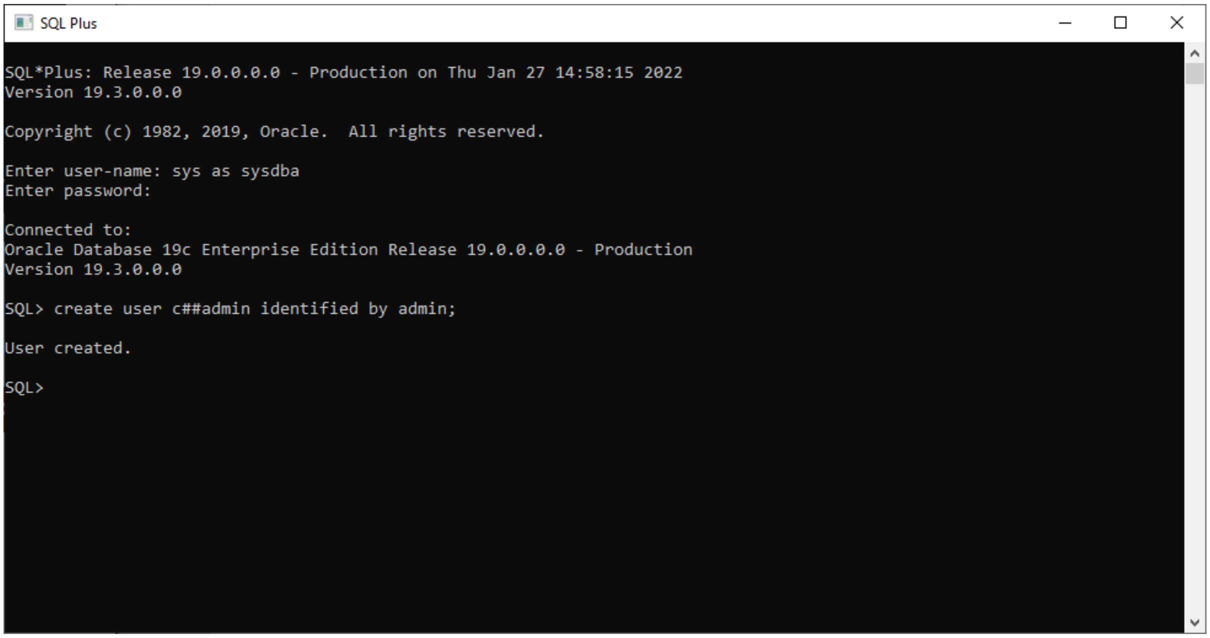
<tr><td>7</td><td>Grant firewall permission to the application by selecting one of the networks by clicking “Allow acess”</td><td>
</td></tr>
<tr><td>8</td><td colspan="2">Give it some time to finish install all the components and like that the application was successfully installed.</td></tr>
</table>

Create an Admin user for database administration

Start → Type “*SQL Plus”* and run application”

Type in the terminal alongside user-name the command “sys as sysdba” follow by an enter press and press enter once again for password which is not require currently.



Note: In my case in named the user “c##admin”

Created an admin user by typing in the terminal commands “create user c##admin identified by admin;”

Grant all the permissions the administration will need by typing the commands as follow:

1.1 “ grant unlimited tablespace to c##admin; ” press enter after

1.2 “ grant resource, connect, dba to c##admin “ press enter and a number 2 in my case came up type “ ; ” beside it then hit enter

1.3 Finally type in “ commit; ” and press enter and should be good with privileges

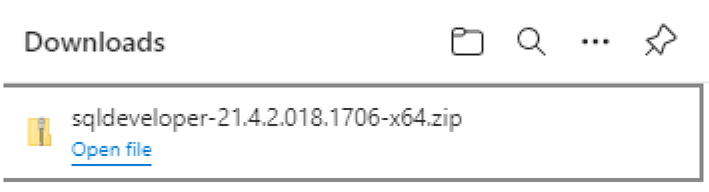
1.4 Type “ exit; “ to close terminal

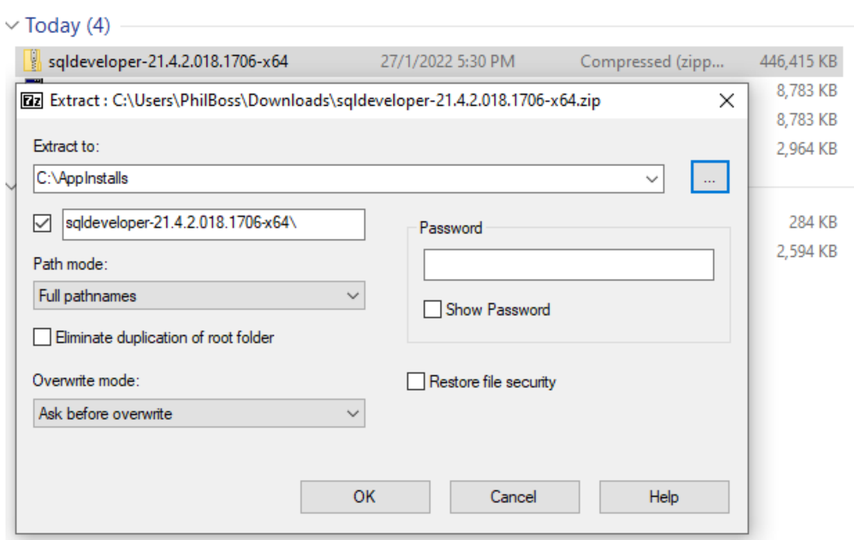
Connect Oracle with SQL Developer

The use of SQL Developer (IDE) to connect to the Oracle database:

Search results for SQL Developer will show the oracle linked software directly. In order to download the software, I had to sign in to my Oracle account. Afterwards the message box below was prompted before downloading zip folder:

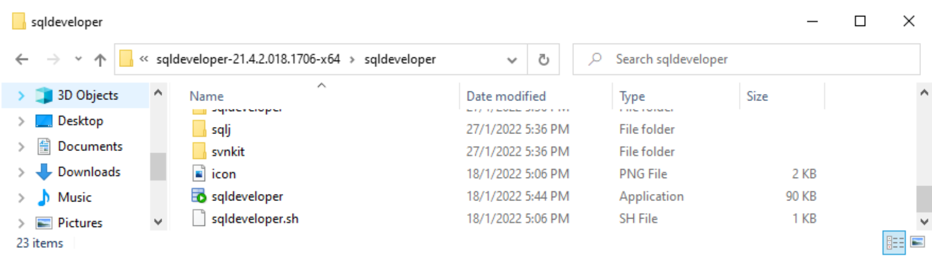
Downloaded the “Windows 64-bit with JDK 8 included” platform version.

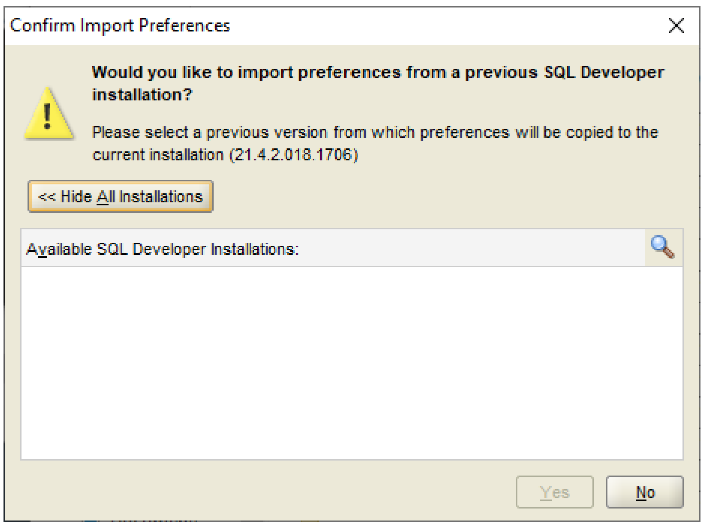




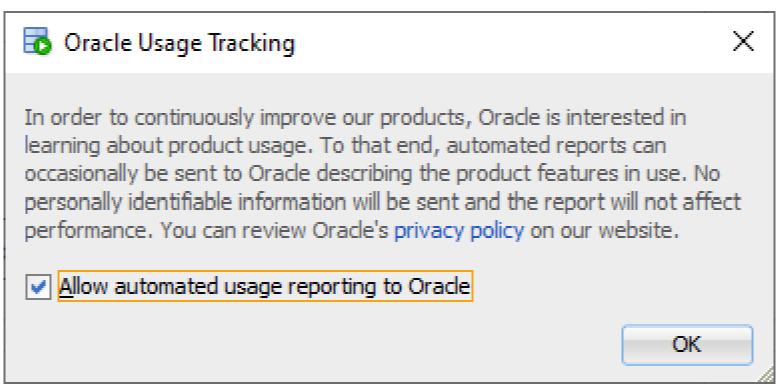
Extract the files to an executable folder in my case I extracted the zip to a folder I created in the root of the drive. C:\

Navigate to the extracted folder and locate the “sqldeveloper.exe” and launch





Clicked NO for the import preferences

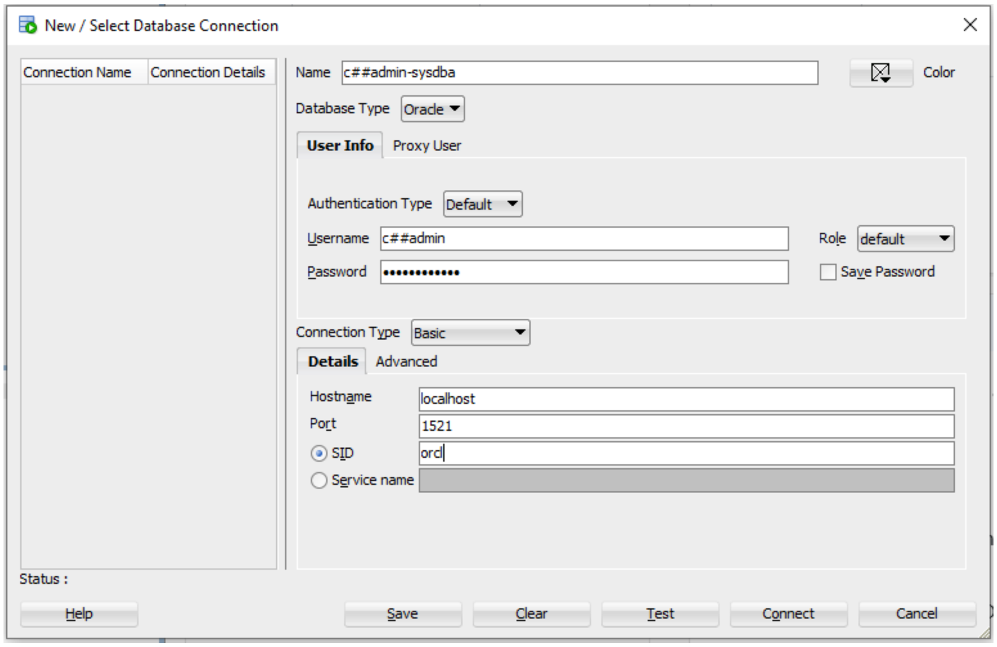


Select OK with the checkbox selected

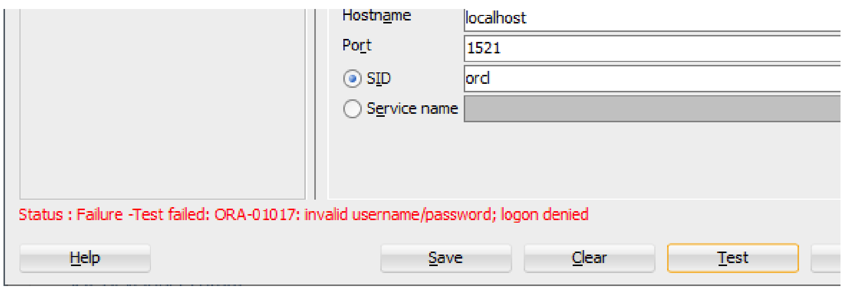
Select “Create a connection Manually” button

- ✓ Enter a connection name of your choice
- ✓ For authentication enter the database username and password corresponding to the user
- ✓ Select “basic” for connection type and hostname “localhost” since hosted on local server

- ✓ Port = 1521 and Sid = orcl



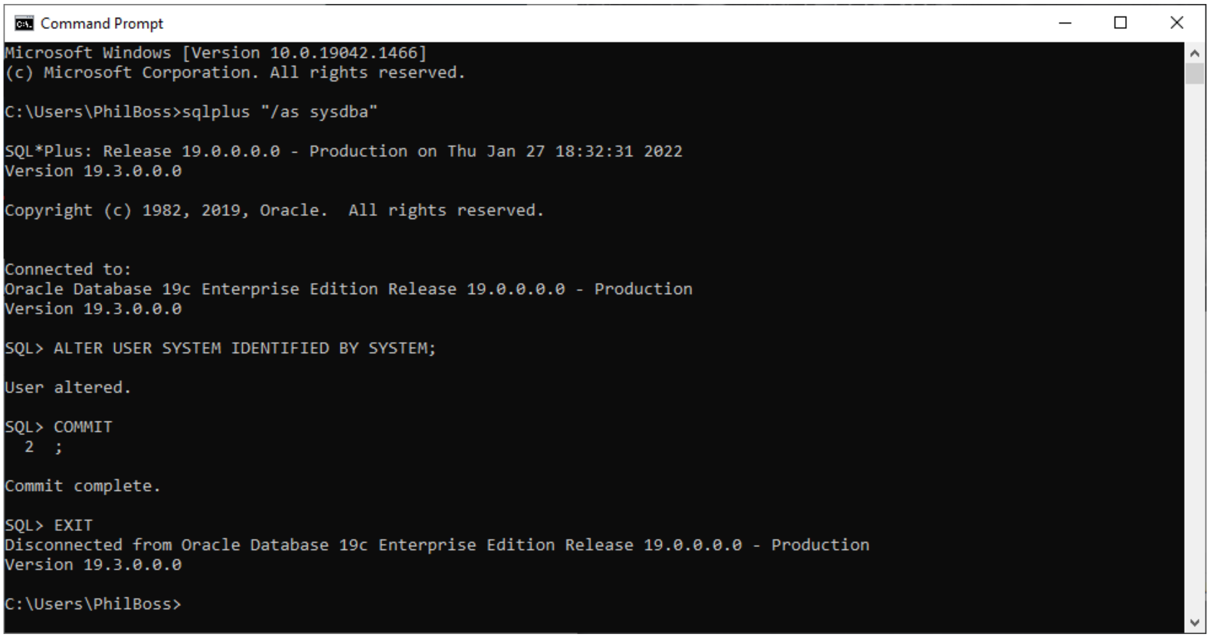
Tested the connection and got an error : Status : Failure - Test Failed: ORA-01017: invalid username/password; logon denied



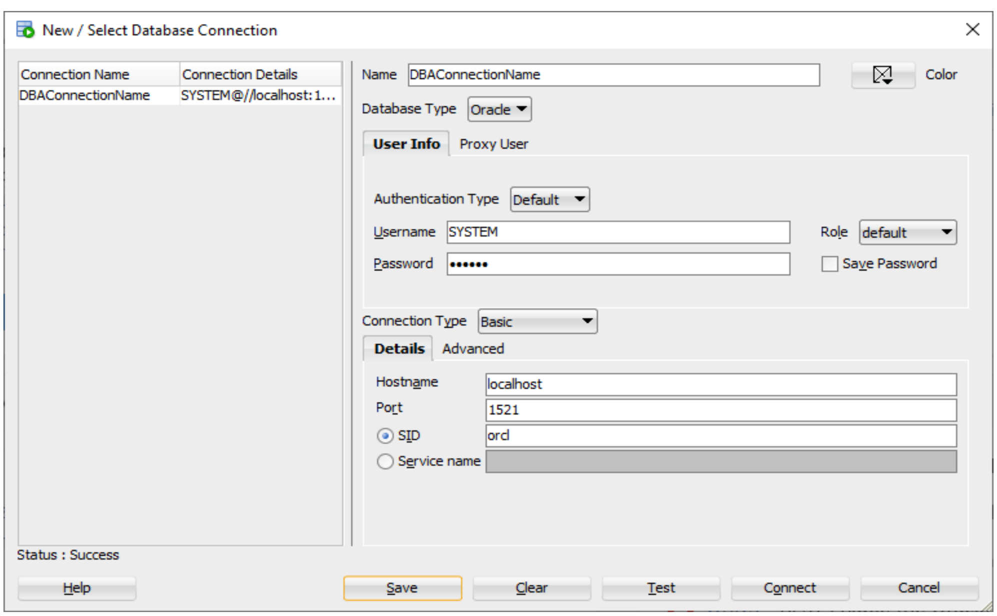
To resolve this error, I search "command prompt" in windows button search and ran the application. Then typed and run the following commands below:

- ✓ sqlplus "/as sysdba"
- ✓ ALTER USER SYSTEM IDENTIFIED BY SYSTEM;
- ✓ COMMIT
- ✓ ;

- ✓ EXIT

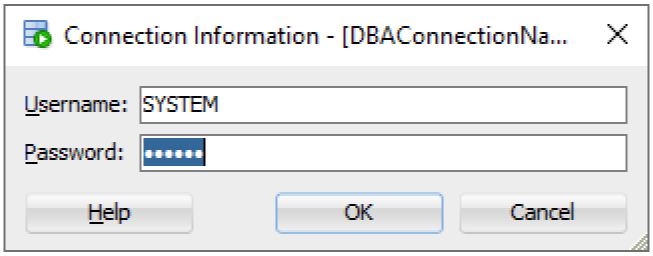


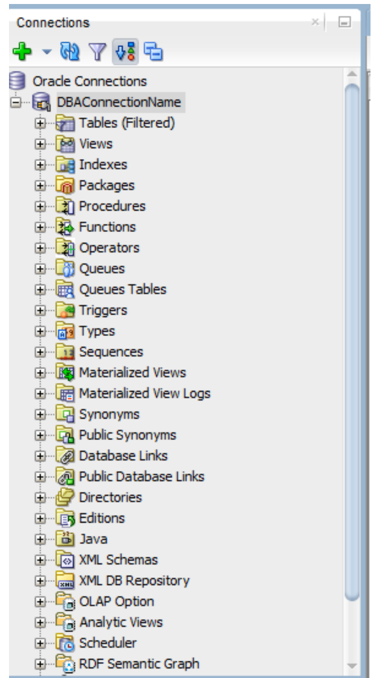
New connection *DBAconnectionName* is created as shown below:



Connection Status was successful

Enter username and password again for connection.





Shutdown Oracle database instance:

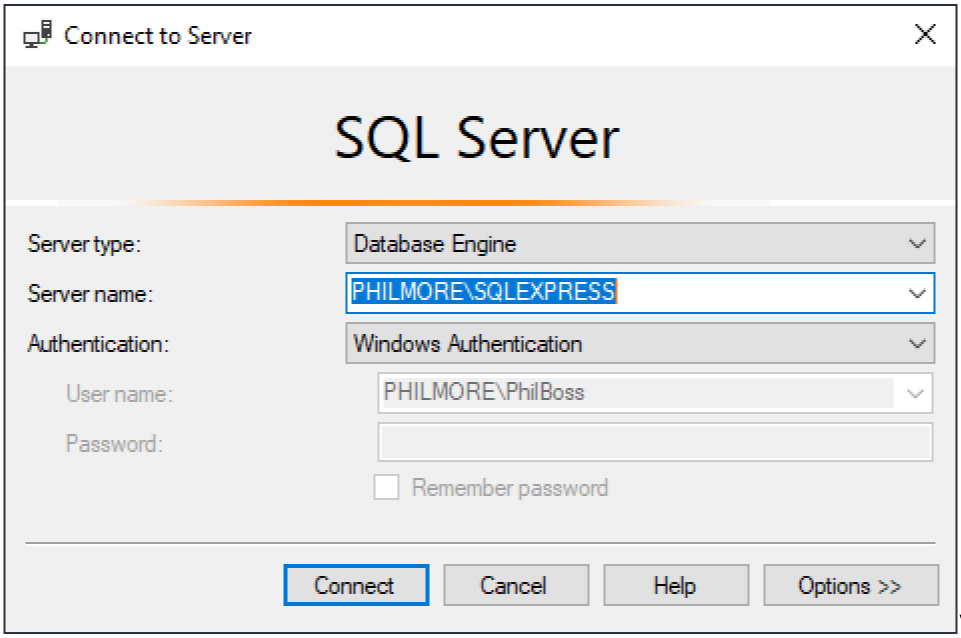
```
SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
```

Startup Oracle database instance:

```
SQL> startup
ORACLE instance started.

Total System Global Area 5133824504 bytes
Fixed Size                  9277944 bytes
Variable Size             956301312 bytes
Database Buffers         4160749568 bytes
Redo Buffers                7495680 bytes
Database mounted.
Database opened.
SQL>
SQL>
```

The **Microsoft SQL Server Installation, Configuration and Connection** part of the lab was already done, as I already had installed in recent months, due to courses done. The snippet below shows the Microsoft SQL Server Instance when launched:



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